Similarity by Selected Traits

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Similarity by Selected Traits before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Repair Bias in Data

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Missing Data

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Choose Informative Questions

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Similarity by Selected Traits trace and record the first step you can explain without help.

Use W2: Compute **distance**s for **Distance** between **feature vector**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Similarity by Selected Traits explanation against the supplied facts before discussing it.

Use the worked case for **Distance** between **feature vector**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Rank independently

Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate **distance**s, select the closest, and identify a tie among the others.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Similarity by Selected Traits target to a trace and a justified conclusion.

Represent books as [animal,short,illustrated] with 1=yes. Query=[1,1,0]; A=[1,0,0], B=[1,1,1], C=[0,0,0]. Count mismatches. What changes if illustrations matter twice as much?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Similarity by Selected Traits case and show the deciding evidence.

Traits are [animals,adventure]. A=[1,1], B=[1,0], C=[0,0]; query=[1,0].

Use number of differing traits as **distance**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

Score **similarity** using one point for same topic and one for same difficulty. Target is space, easy. A is space, hard; B is animals, easy. What happens?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 21 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Similarity by Selected Traits adds to Transparent Book Recommender and keep the evidence.

Make selected traits and their **scale**s visible in every recommendation explanation.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Represent books as [animal,short,illustrated] with 1=yes. Query=[1,1,0]; A=[1,0,0], B=[1,1,1], C=[0,0,0]. Count mismatches. What changes if illustrations matter twice as much?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Read a vector

> Read a vector. Apply the supplied Similarity by Selected Traits facts and keep your reasoning.

For [4, 1] under the worked feature order, identify energy and group-work ratings.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Compute distances

> Compute distances. Apply the supplied Similarity by Selected Traits facts and keep your reasoning.

From target [1, 2], find **distance**s to A [2, 2] and B [0, 4].

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Repair feature order

Repair feature order. Apply the supplied Similarity by Selected Traits facts and keep your reasoning.

One record uses [group work, energy] while others use [energy, group work]. Why is the **distance** unreliable?

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Rank independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Rank independently. Apply the supplied Similarity

> by Selected Traits facts and keep your reasoning.

Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate **distance**s, select the closest, and identify a tie among the others.

My reasoning and evidence

W5 Limit similarity

> Limit similarity. Apply the supplied Similarity by Selected Traits facts and keep your reasoning.

Two activities have **distance** zero. Must they suit every learner equally well?

My reasoning and evidence

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Similarity by Selected Traits

> A fictional toy book matcher represents two preferences with numeric features. Explain the labelled step before checking the changed condition.

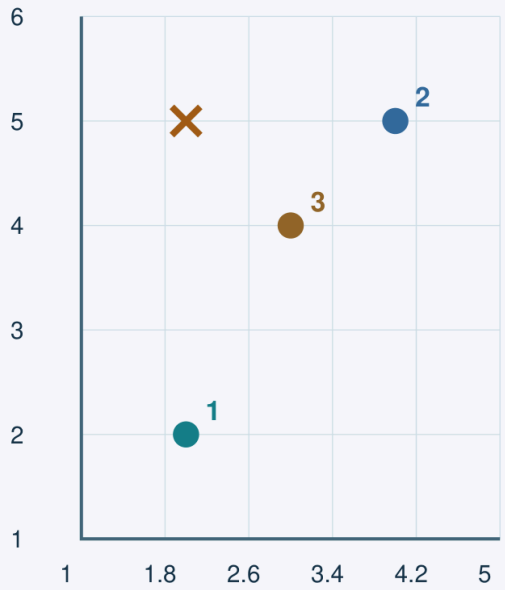
A fictional toy book matcher represents two preferences with numeric features.

Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

Use **Manhattan distance**: add absolute **coordinate** differences; return all smallest-**distance** candidates.

AI ACADEMY / AI-2/21

Toy feature positions



Horizontal: first toy feature; vertical: second toy feature

1: A at (2, 2)

2: B at (4, 5)

3: C at (3, 4)

Cross: Target at (2, 5)

Distance rule: Manhattan distance; horizontal=first feature, vertical=second feature

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/21

Similarity by Selected Traits

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Distance A

$|2-2|+|5-2|=0+3=3$.

↓

2. Distance B

$|2-4|+|5-5|=2+0=2$.

↓

3. Distance C

$|2-3|+|5-4|=1+1=2$.

↓

4. Choose closest

B and C tie at distance 2; keep the tie visible.

CHANGE ONE THING

Give the first-coordinate difference weight 2 while the second keeps weight 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why B ceases to be a closest option without any candidate data changing.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional toy book matcher represents two preferences with numeric features.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B and C tie at **distance** 2; keep the tie visible.

1. **Distance** C

2. Choose closest

3. **Distance** A

4. **Distance** B

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Choose closest B and C tie at **distance** 2; keep the tie visible. Each **coordinate** contributes according to the declared **distance** rule.

Smaller **distance** means closer in this representation, not automatically a better reading experience. If one **coordinate** uses a much larger numerical **scale**, it can dominate the sum even when it is not more important.

Feature units and scaling belong in the method record.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Give the first-**coordinate** difference weight 2 while the second keeps weight 1. Which conclusion is supported by the supplied facts?

1. **Distance**s become A=3, B=4, C=3; A and C tie.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Distance**s become A=3, B=4, C=3; A and C tie. The representation's comparison rule changes; the supplied vectors stay the same.

Repair a missing building block

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?.

Return to this week and explain what you changed.

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Choose the best original question and explain why its more balanced split matters here..

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Similarity by Selected Traits picture from its labels, then record a relevant limit.

01. Similarity by Selected Traits: the case

AI ACADEMY / AI-2/21 · 01 · LOOK AT THE FACTS

Similarity by Selected Traits: the case

The situation

A fictional toy book matcher represents two preferences with numeric features.

Supplied evidence

Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].
Use Manhattan distance: add absolute coordinate differences; return all smallest-distance candidates.

By the end, I can calculate a feature distance and state how scaling changes its meaning.

Follow the labelled picture with the matching explanation. Open the vector diagram

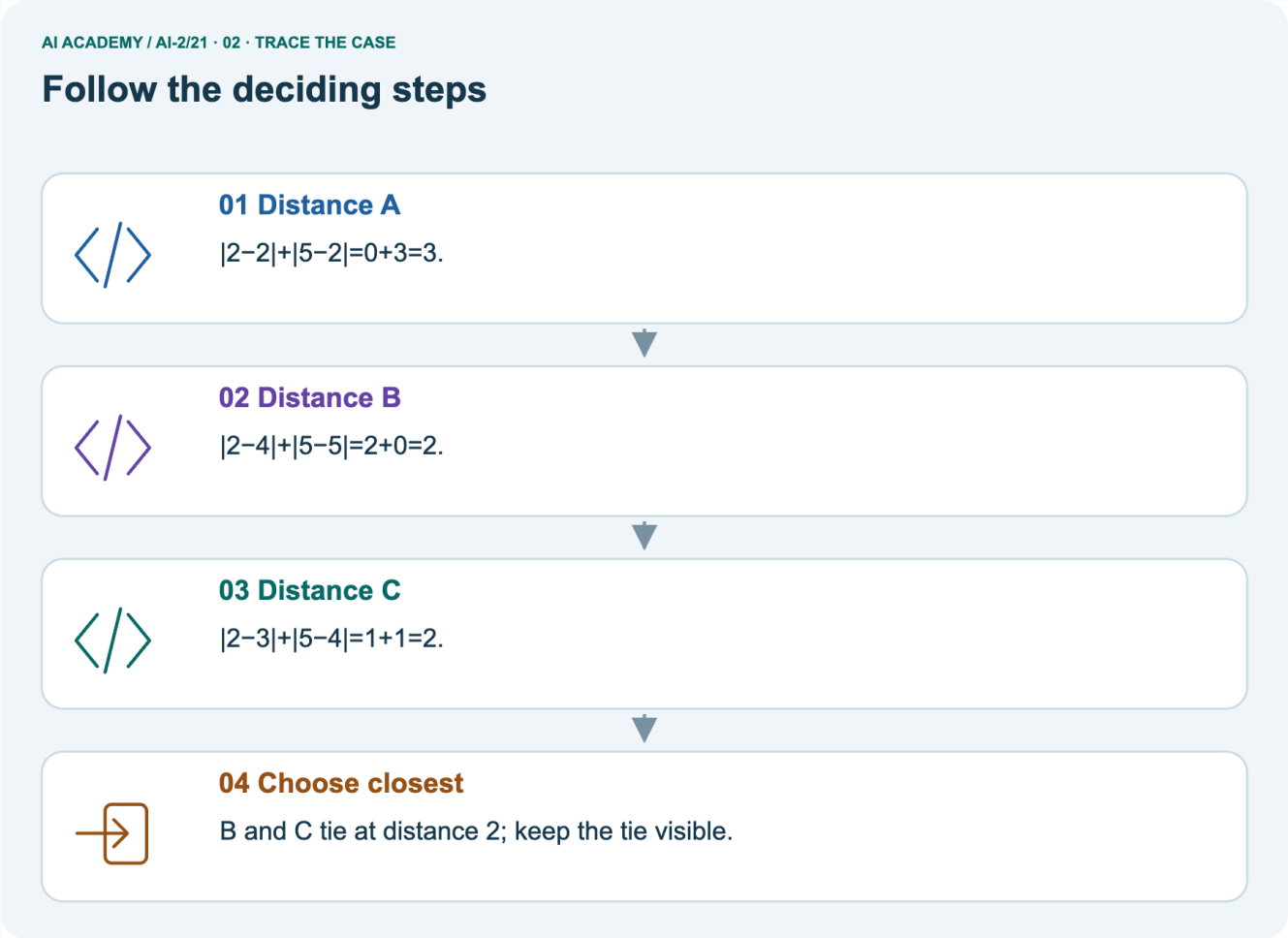
A fictional toy book matcher represents two preferences with numeric features. Supplied facts: Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

Use **Manhattan distance**: add absolute **coordinate** differences; return all smallest-**distance** candidates.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



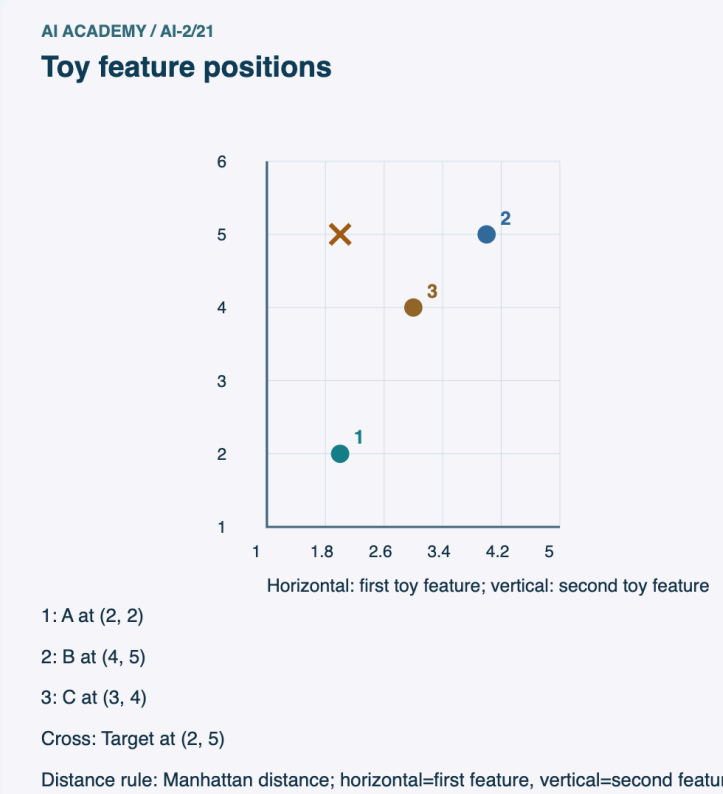
Follow the labelled picture with the matching explanation. Open the vector diagram

Distance A: |2−2|+|5−2|=0+3=3. → **Distance** B: |2−4|+|5−5|=2+0=2. → **Distance** C: |2−3|+|5−4|=1+1=2. → Choose closest: B and C tie at **distance** 2; keep the tie visible.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Toy feature positions



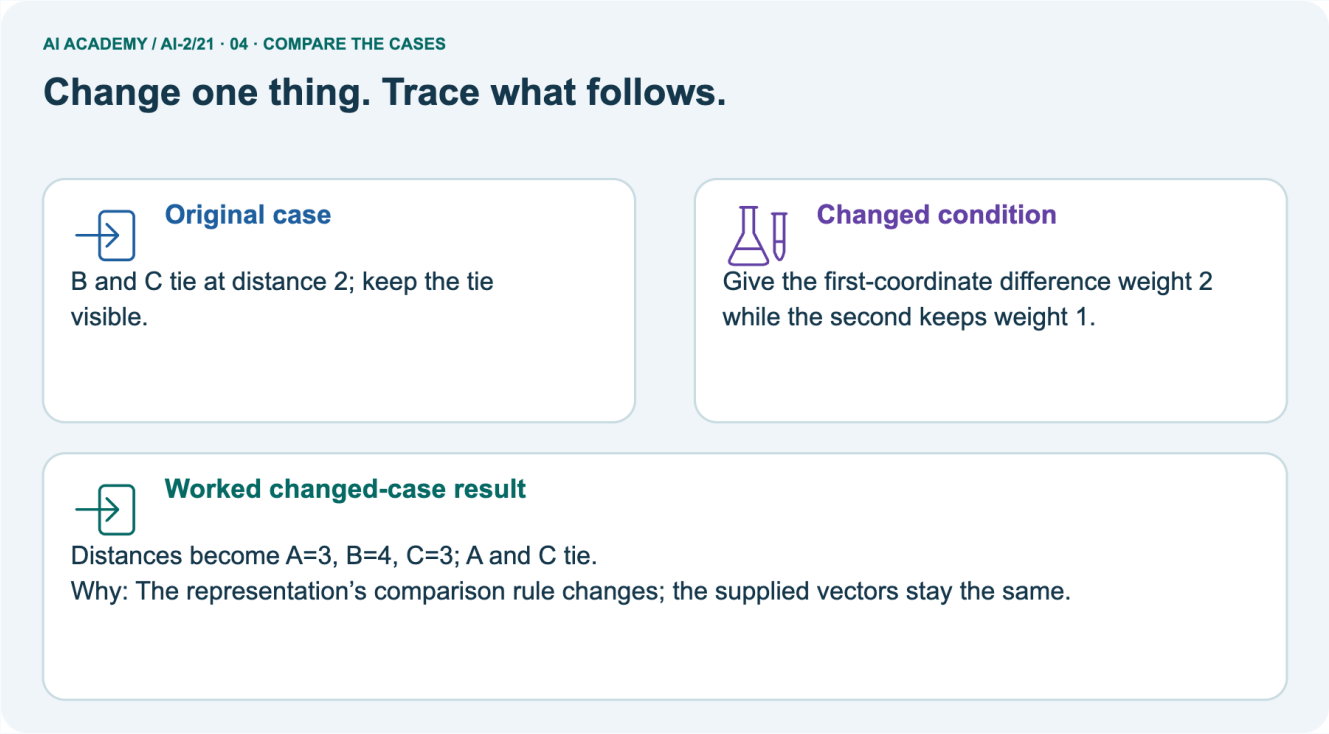
Follow the labelled picture with the matching explanation. Open the vector diagram

Toy feature positions. A: (2, 2). B: (4, 5). C: (3, 4). Query Target: (2, 5). Axes: first toy feature; second toy feature. **Distance**: **Manhattan distance**; horizontal=first feature, vertical=second feature.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

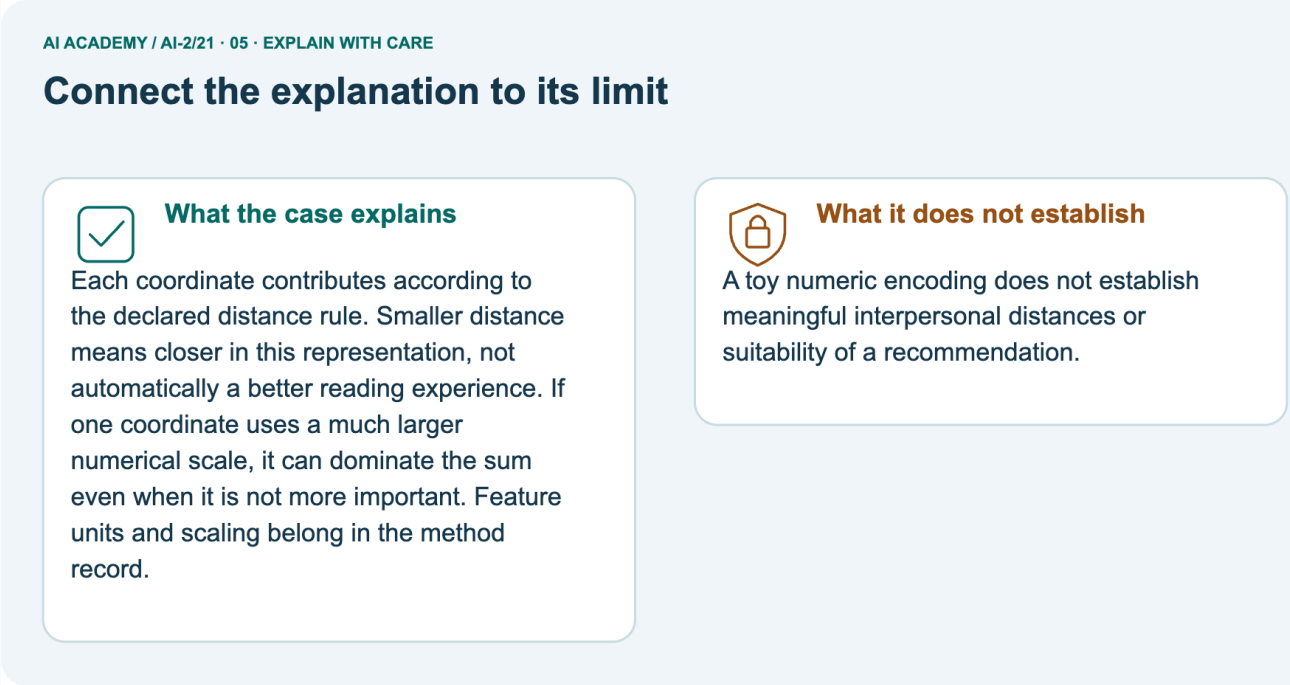
Original case: B and C tie at **distance** 2; keep the tie visible. Changed condition: Give the first-**coordinate** difference weight 2 while the second keeps weight 1. Expected result: **Distance**s become A=3, B=4, C=3; A and C tie.

Reason: The representation's comparison rule changes; the supplied vectors stay the same.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Each **coordinate** contributes according to the declared **distance** rule. Smaller **distance** means closer in this representation, not automatically a better reading experience.

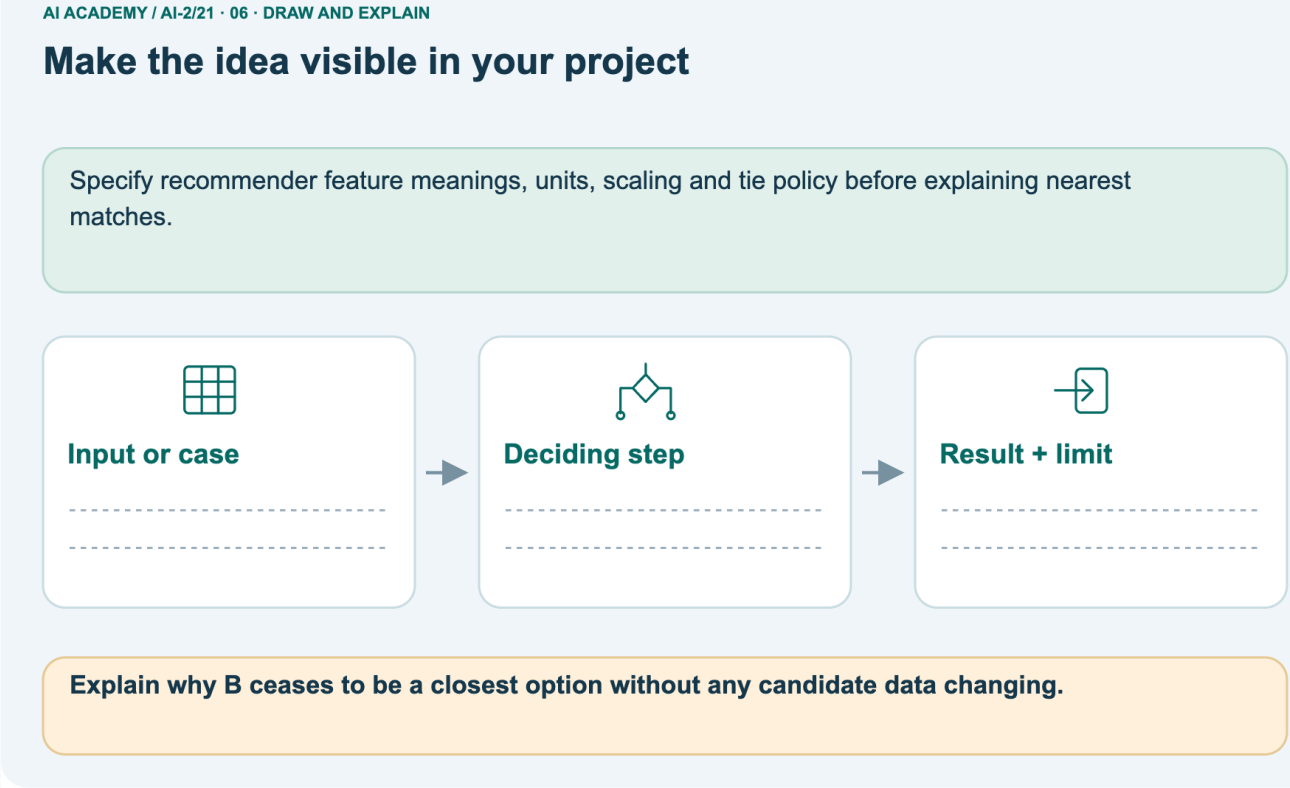
If one **coordinate** uses a much larger numerical **scale**, it can dominate the sum even when it is not more important. Feature units and scaling belong in the method record.

Limit: A toy numeric encoding does not establish meaningful interpersonal **distance**s or suitability of a recommendation.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B ceases to be a closest option without any candidate data changing.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.